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Manus AI: The Butterfly Effect Technology (A)

It was a Monday morning in late December 2025, and Tao "Hidecloud" Zhang, cofounder and chief product officer of Butterfly Effect Technologies had just landed in Singapore, returning from a trip to the Bay Area. He was tired after the long journey, but thankfully Butterfly Effect's Manus agent had already pre-processed his calendar, prepared briefing packets, customized the necessary slide decks, and drafted research summaries to provide deep background on the companies represented in the day's meetings. Before the meetings started, Tao went to look for his cofounders Xiao "Red" Hong, Butterfly Effect's cofounder and CEO, and Ji "Peak" Yichao, chief scientist and fellow cofounder. He wanted to debrief them on some interesting discussions he had had while in the U.S.

Only nine short months ago, on March 5, 2025, the team had launched Manus. Their vision was clear. "The brain already exists; what's missing is the hand," Red liked to say as he debated the concept with Tao and Peak. In contrast to large language models (LLMs)—the powerful artificial intelligence (AI) systems built on deep learning and trained on massive amounts of text and data to understand, generate, and process human language for tasks like answering questions, summarizing, translating, and writing—Manus was all about building that hand: an AI system that went beyond responding to prompts by organizing the work and operating continuously to execute complex tasks and augment human capability.

Once caught up on the West Coast conversations, Tao went back to his desk, still contemplating the same old debate. Manus was targeting consumers (or pro-sumers),[a] and uptake had been phenomenal. Moreover, the team had learned about consumer behaviors, needs, and use cases, growing by leaps and bounds. Recently, they had crossed $100 million in annual recurring revenue (ARR), making the company the fastest in the world to grow from $0 to $100 million. Most outsiders, however, believed that competition for consumers was going to be fierce: the larger LLM players would undercut the Manus business model and were already putting pressure on Manus pricing. The only option, the team argued, would be a shift to the enterprise market to maintain and scale revenues for the long haul.

The debate was becoming existential. As Google launched its Gemini 3 models and other AI products, marking significant advancements in capabilities and revealing a growing strength in agentic

[a] Pro-sumers referred to consumers with professional needs who purchased instances for their own use.

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AI, analysts questioned the long-term sustainability of independent AI agent startups.[1] Further, as former Google CEO Eric Schmidt had noted, "Chinese startups literally cannot get the money to keep up in the AI race, lacking access to the depth of the U.S. capital markets."[2] Operating an agent business was an expensive proposition; Manus to date had processed close to 150 trillion tokens,[b] which necessitated costly application programming interface (API) calls to a variety of foundation models. Some large U.S. frontier lab[c] players were already knocking at Butterfly Effect's door. This very topic had been the subject of several conversations Tao had just had on his latest trip.

The Context: Artificial Intelligence and Agents

Agents: How They Work, Play, and Pay

An agent was a software program that exhibited "agency"—the ability to make decisions and act. Pragmatically, an agent was AI that handled multi-step tasks without a human "steering" or prompting it.[3] It was a software system that could perceive, reason, and act within an environment to achieve specified goals. An agent took in information, or inputs. Its policy or reasoning process, determined by its architecture, design, and training, helped it decide what to do and perform actions based on those decisions. It learned by doing. Some examples included autonomous robots, chatbots, game-playing systems, and workflow automation agents. (See **Exhibit 1** for additional information on building an agent.)

Since 2023, agents had evolved quickly and incorporated memory systems allowing them to recall prior interactions and user preferences. Increasingly, a level of tool agnosticism and cross-brand integration allowed agents to choose which foundation model might best solve problems. They could work for longer periods of time on a single task; developers using Anthropic's latest model, Claude 4, reported it could work for up to seven hours on a coding problem.[4]

Agents could take multiple actions, often by connecting to other applications, like customer relationship management systems (CRMs), coding software, or repositories like GitHub to complete tasks.[5] Cursor customers using its coding agent feature, for example, could turn to bug tracker Jira for information about software bugs. Anthropic's open-source software Model Context Protocol (MCP) helped AI models access this information too.[6] Agents could mix and match LLMs with reasoning models or other planning components to determine which steps to take in what order to complete a task.[7] An "orchestrator" software coordinated these different technologies. MCPs opened the doors for agents to connect to and use applications or draw on information from applications.

By July 2025, various types of AI agents had proliferated (see **Exhibit 2**). They appeared as stand-alone apps, as sidebars or add-ons in existing apps, or as dialog boxes on webpages.[8] Microsoft, Salesforce, SAP, and others offered agents that automated white collar tasks in finance, human resources, and sales.[9] Companies developed agents tailored to their needs; an online car retailer built an agent that allowed employees to build new webpages by describing how the webpage should look in plain English.[10] More advanced capabilities quickly followed; Microsoft developed a site reliability engineer (SRE) agent that monitored a customer's webpage 24/7, tracking if it went down or acted buggy and resolving issues with patches without human input.[11]

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[b] Tokens in this instance referred to words, or parts of words.

[c] Frontier labs were the leading, deep-pocketed, cutting-edge organizations developing the most advanced and powerful AI models. In late 2025, they included Anthropic, Google DeepMind, Meta, and OpenAI.

Pricing for agents varied, and consumers typically paid a premium over the cost of ChatGPT's most expensive tier, about $200 a month.[12] While OpenAI had suggested users would pay $20,000 a month for its PhD-level research agent,[13] by 2025, pricing was a bit more reasonable. OpenAI's Codex coding agent cost $6 per 1 million output tokens, about four times what it charged for access to ChatGPT's LLM.[14] Users of Microsoft's coding agent features—accessed through GitHub—paid about $40 a month, again about four times the price of its more basic coding tools.[15]

While the cost of running an agent was "a significant step up," as one executive said, it brought greater value than reliance on basic ChatGPT.[16] One firm reported agents had helped it reduce its reliance on software engineers, and saved it "hundreds of thousands of dollars annually."[17] A large consulting firm estimated that it saved 5% to 20% using an agent that automated compliance audits of customers' IT rather than relying on humans to manage the risk analysis manually.[18]

The Agentic AI Landscape in 2025

Given these developments, by 2025, the agentic AI sector had seen dramatic developments. The earliest successes were specialized, developer-facing coding assistants, like Cursor and Windsurf, that were maturing into full engineering platforms. Following a different trajectory were foundation model vendors, such as frontier labs OpenAI and Google, that shipped agentic capabilities directly into existing user workflows. These might include cloud virtual machines, browser automation, and long-horizon control. (See **Exhibit 3**.)

The year had seen AI coding agents report dramatic growth and investments; in November, Cursor closed a $2.3 billion Series D at about a $29.3 billion valuation.[19] Windsurf rode out several waves of drama, including an unsuccessful OpenAI acquisition bid, with the company's core team moving in July to Google for about $2.4 billion,[20] while Cognition acquired its assets for an undisclosed amount.[21]

Engineers and premium enterprise customers quickly adopted the developer integrated development environment (IDE)/agent offerings. Their popularity kicked off a massive war for talent, threatening many startups' viability. In 2024, Google invested $3 billion in startup Character AI, and in June 2025, Meta invested $14.3 billion in startup Scale AI.[22] These deals were increasingly structured as licensing arrangements that included talent moving to the purchaser, rather than outright acquisitions.[23] A team of former Google developers, who had left to create chatbot start-up Character.AI, rejoined the firm in such a deal in 2024.[24] Likewise, Microsoft hired most of AI startup Inflection's team—including its founder—in a March 2025 licensing agreement valued at over $650 million, and startup Adept licensed its technology to Amazon for at least $330 million, with many of its AI researchers moving to the tech giant.[25] Observers believed such deal structures evaded regulatory scrutiny. Other types of deals proliferated as well, including those through which large players like NVIDIA and Oracle provided AI companies with chips and cloud server technology as investments.

Agents Miss the Mark?

Companies like OpenAI and Salesforce had hailed 2025 as the "year of the agents," sparking a full-blown fear that fully autonomous AI would quickly take over human jobs. Others, however, expressed skepticism, suggesting a long horizon lay ahead before AI could replace humans in a host of roles. Said an OpenAI founder and leading AI researcher Andrej Karpathy, "Overall the models are not there. [. . .]. I feel like the industry is making too big of a jump and is trying to pretend like this is amazing, and it's not. It's slop."[26] Some companies acknowledged use of agents, and had even seen ways to cut staff, but found they then needed new staff with different skills to monitor agents' work. One ecommerce marketplace app CEO said, "I think it's interesting I can replace some $400,000-a-year jobs [with agents], but I can't completely replace them. I just have someone else I pay $400,000 to manage agents to become four times as productive and then fire three people."[27] He added, "AI-generated code needs to be [checked] as the product experience can't be validated by the AIs yet, and that's a different task from coding."[28]

While agents excelled at coding and customer support tasks, they fared poorly on tasks without clear right or wrong answers, including generating slide decks. ChatGPT's and Gemini's LLMs could give instructions on how to complete tasks like resume-screening and financial documents or stock analysis, but they could not complete the tasks themselves without human direction.[29] And while the LLMs could surface the relevant research that had solved difficult math problems, they were unable to correctly solve them themselves.[30]

Nightingale

Manus's future team members met in their first year at Huazhong University of Science and Technology in Wuhan, China, where they came together as members of Unique Studio, an elite student engineering club. There, they thrived in an environment that was intense and pragmatic: learning by building real products, often under severe time constraints. Chuangshen ("PanPan") Pan, Manus's chief technology officer and fourth cofounder, recalled, "If you didn't want to ship, you didn't last."

After graduating in 2015, Red formed his first startup, working on a social app. "We did a lot of things wrong," PanPan recalled. "But it taught us how hard it is to build something people will use every day." Not long after, Red won a hackathon where he met Liu Yuan, a general partner at ZhenFund, who became his mentor and close friend. Yuan would follow Red for the next decade, funding his efforts, and—importantly—introducing him to Peak and Tao.[31]

Under Red's direction, the small team pivoted to a new idea—building a browser extension to manage WeChat business accounts. The tool, Yiban Assistant, provided an efficient browser-based editor and plugin for managing WeChat Official Accounts, one of few official extensions in a sea of unofficial automation tools. It helped users quickly format articles and manage content for WeChat public accounts and supported functions like inserting styles, editing images, finding images and emojis, and collecting materials. Available as a browser plugin, it incorporated AI-powered writing features for generating content, optimizing titles, and performing content diagnostics, and offered data analysis and batch message reply capabilities.

Yet, the tool lacked a real monetization strategy, and growth was a challenge. However, when WeChat cracked down on unofficial tools and banned illegal API use, the extension went viral as companies scrambled to find compliant alternatives. Registrations on Yiban quickly surged into the millions of users. Monetization remained a challenge, however. A buyer offered to acquire Yiban, and the founders sold 51% of the company, assuming they could stay on and continue to develop the technology. Soon they discovered that without ownership control, they were unable to innovate and build what or how they wanted. They left the company.

A Butterfly Effect

In 2022, just as ChatGPT launched, Red founded Butterfly Effect Technology with PanPan and fellow classmate Henry Yang, who was chief marketing officer.

Red, Peak, and the small team began experimenting with AI and soon launched an AI-powered browser extension called Monica.im. The product—essentially acting as a "wrapper" for underlying frontier models like ChatGPT and Claude—aggregated across multiple LLMs. It offered a unified interface for users to translate, summarize, and rewrite content on any webpage. Users flocked to it, quickly growing the base to over 10 million. Monica was soon profitable, giving Hong and the team the freedom to stay independent, including staving off a $30 million acquisition offer from ByteDance.[32] However, the team observed limitations with Monica: most users did not understand browser extensions, and the interface constrained autonomy.

An AI Browser?

In March 2024, Tao and Peak joined as cofounders; Tao was head of product, and Peak took the role of chief scientist, leading technical and infrastructure development. CZ joined as chief operating officer. Tao had 15 years' experience in the industry, primarily involved in startups with a sharp focus on product, including at Wandoujia and Sensordata. Before joining the Manus team, he had been head of global product at ByteDance and held product roles at Tencent. CZ had a business background; as she said, "In the founder team, I am the only one that cannot code." She had overseen cross-border acquisitions and worked in China's largest real estate firm. Peak, a high-school dropout and a long-time open-source contributor, had shipped his first software product while in high school. He developed jigsaw, a data visualization toolset, and Mammoth, a third-party iPhone browser that went viral in China and won him the Macworld Asia Grand Prize in 2011. At the age of 20, in 2013, he was named to *Forbes'* China 30 Under 30 List.[33] He founded Peak Labs, where he developed Magi, an ambitious search engine designed to extract structured answers directly from the web.[34] Built in the pre-foundation-model era, Peak's team trained Magi on custom models; as he noted, "It was before ChatGPT. The natural choice was to make your own model." However, these were expensive and slow to improve and were quickly eclipsed by ChatGPT-3, Flan-T5, and other large-scale models, making his in-house models "irrelevant overnight."[35] Peak recalled, "It was an 'aha' moment, but also doomsday for me." He sold his startup to a Chinese unicorn company and was named to the *MIT Technology Review*'s Innovators Under 35 list.[36]

The team's experience building Monica inclined them to consider the browser as a way in to building a product that was able to inference.[d] Peak said, "There's lots of things ChatGPT does not know about me, but the browser knows me." Tao added, "Inference is based on all the context of everything you are reading, seeing, listening to, etc."

They soon realized: "The problem is not the model; it's us." Tao said, "The model is already smarter than most humans. You have to ask a question to get an answer. The real limitation is the capability and motivation to ask the question. The chatbot needs you to ask a lot of questions, then you can get the value of the model. Many people lack this capability and motivation." The browser approach offered a potential tool for the models to move past waiting for questions to be asked. The team studied tools like Perplexity AI, and The Browser Company of New York. The browser was where work happened, one observer noted, adding, "But today's browsers were designed for browsing, not working."[37] Tao recalled, "For the last two and a half years, the biggest problem in the industry is that everyone is building the brain. We want to build the hand. No one else is building the hand."

The team worked feverishly on their idea, assigning 15 of their 35 employees to the project. Tao acknowledged, "It was a big bet for a small company like us. We spent March to October working on

[d] Inference, in AI terminology, meant that the AI model was actively thinking, reasoning, and processing information to accomplish its tasks.

it." By summer 2024, the project appeared to converge, but as it was getting close, and the team worked on the prototypes, they started to have increasingly serious doubts. Tao recalled:

> The experience was kind of weird. The browser lived on your computer. The AI starts to do stuff, like browse a LinkedIn profile, and mark a potential candidate. And while it's doing that, you can't move your mouse or use your keyboard because that would break the AI's work. It's like you hired an intern but didn't give them a computer, so you have to share yours with them. You're competing for the mouse. It's very weird.

They killed the project. The experience was humbling, Tao acknowledged. "It was the first project I led that failed," he noted. But it made clear that autonomy required separation: an agent would need its own environment. The 12- to 15-person team now had time on their hands. "We had nothing to do for three weeks," Tao said. "We just explored new ideas, trying different demos." Some of the team and their family members—who were not coders—started to use Cursor to solve daily tasks. Tao noted, "My wife has no coding background. She used Cursor to generate charts from raw data. We saw an opportunity there: normal or average people could benefit from the power of AI coding." Their insight: coding was not the bottleneck; rather, the interface was.

Building Manus

If model progress is the rising tide, we want Manus to be the boat, not the pillar stuck to the seabed.

— Peak Ji[38]

By November 2024, the team started to work on a new tool, building off their experience with the abandoned AI browser. The new ambition was to build an agent with its own dedicated browser, capable of doing its own research, finding all the necessary information and content, and getting the work done. Tao said, "You can just use it right away. That's the basic concept." Borrowing inspiration from Cursor, they used a similar conversation panel that showed what the AI was thinking, and what its next step would be. They designed the tool to let the AI run on the cloud instead of on the user's computer. Tao explained, "Each task in Manus is assigned its own virtual machine. Let the AI use its own computer, not your computer." They landed on the name Manus, drawing on the Massachusetts Institute of Technology's motto "*Mens et Manus,*" or mind and hand. Tao added, "All these frontier models, they are just the brain, super smart. But as humans, we can't make an impact in the physical world without using our hands to touch and build." Manus would infer intent and work continuously. Tao said, "The agent might as well be working." That winter, the team closed another funding round, this time with ZhenFund, HongShan (HSG), and Tencent, valuing the company at almost $100 million.

The team decided not to train its own models, instead building Manus to be agnostic across models, choosing the best available one to understand, structure, and execute any number of tasks. Peak leaned on context engineering, citing the "bitter lesson" he had learned in his last startup, where he trained models from scratch.[39] Context engineering structured the information provided to a foundation model to elicit the desired behavior. Yet context engineering was an experimental science. Peak acknowledged, "It turned out to be anything but straightforward." The team rebuilt the model four times, each time discovering a better way to shape context. Peak added, "We affectionately refer to this manual process of architecture searching, prompt fiddling, and empirical guesswork as 'Stochastic

Graduate Descent.' It's not elegant, but it works."[40] Context engineering would help the team focus on what to include in the event stream[e] and how to compress content. (See **Exhibit 4.**)

The core challenge in building Manus was managing the immense computational cost of an autonomous agent. Unlike a standard chatbot that responded once to a single question, an agent like Manus operated in a continuous loop, often consuming 1,000 times more tokens to complete a single complex project. To make this economically viable, the team focused on a metric called the KV cache hit rate,[f] which essentially measured the system's ability to "reuse" its past work (see **Exhibit 5**).[41] In business terms, this was akin to a consultant leveraging a library of past slide templates rather than building every deck from scratch; by successfully reusing previously computed data, Manus could minimize time-to-first-token[g] and reduce costs by as much as 90%. As Peak noted, "Using state-of-the-art models like Claude Sonnet, this reuse could drive costs down from $3 per million tokens to just $0.30 in some cases, representing a massive leap in unit economics." Costs from $1 to even $70 per token were common.

Beyond raw efficiency, the team developed additional bespoke capabilities for repetitive tasks. By observing user behavior and identifying common failure patterns, the team introduced atomic capabilities—simple, modular tools such as custom PDF generators or voice converters—that the agent could use to navigate obstacles. These tools followed a "Unix philosophy," performing a single task with high reliability so they could be modularly chained together into sophisticated, multi-step workflows.

However, as the number of capabilities increased, the team had to ensure that the agent remained focused and reliable throughout long-horizon tasks by employing a strategy known as masking**.** In a typical AI interaction, the model would be presented with a vast action space—a menu of all available tools and commands. However, having too many choices could lead to distraction or errors in logic. To maintain stability, Manus used a context-aware system that temporarily masked irrelevant tools based on the current step (see **Exhibit 6**). For instance, if the agent was in the middle of a data analysis step, the system prevented it from accidentally sending an email or browsing an unrelated site. By effectively enforcing the correct choices through this architectural constraint, the team ensured the agentic loop remained stable and directed toward the user's goal.

How Manus Works

My dream is for the agent to inference 24/7 for each user. Right now, we are getting close to that dream.

— Tao Zhang

Going beyond the capabilities of a single model, Manus employed a multi-agent architecture that organized its cognitive processes into specialized modules.[42] This brought the engineering advantages of cost efficiency and performance tuning without billion-dollar training costs. Manus consisted of at least three coordinated agents working together: a planner agent that acted as a strategist; an execution

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[e] An event stream was a continuous, ordered flow of data (events) representing happenings (e.g., user clicks, financial trades) in a system, captured in real-time as immutable records, allowing systems to react immediately rather than batch-process later, forming the backbone of event-driven architecture for insights, fraud detection, and dynamic updates.

[f] KV referred to computed key (K) and value (V) tensors.

[g] Time-to-first-token (or TTFT) was a key AI metric measuring the milliseconds it took from sending a prompt to an LLM to its output of the first piece of text, or token. It was crucial for interactive apps like chatbots because it dictated perceived responsiveness.

agent that took the planner's plan and executed it by calling for the required operations or tools; and a verification agent that acted as quality control, reviewing and verifying the execution agent's outcomes.[43] Manus executed its tasks in a full Ubuntu Linux sandbox (virtual machine), that provided it with a comprehensive set of tools that mirrored those of a human user: a shell terminal, a controllable web browser, a persistent file system, and code interpreters. Its server-side environment was persistent, allowing Manus to work continuously, even without the user, a key differentiator from browser-based agents.

The architecture choice(s) allowed autonomous plan formation, tool use, and result verification. It also enabled scale. For large research tasks (i.e., analyzing hundreds of products or papers), Manus's controller decomposed the task into many subtasks, and could launch a full Manus instance for each one. This could mean hundreds of agents worked in parallel. Results were synthesized by the controller. Eventually, Tao envisioned Manus proactively launching research, reviewing calendars, emails, and documents in the background, and surfacing insights in anticipation of a user's request.

Early testers reported hallucinations, long-tail failures, jittery sequences, and re-planning instability,[h] and the team doubled down to fine-tune the agent. Safeguarding and guardrails were also important. The LLM providers caught many unsafe attempts. They also developed control levels that differed by action type. Manus required manual confirmation to send email, while reading the user's inbox did not. The team discussed sensitivity around ethics and safety on a case-by-case basis.

In contrast to what they saw around them—AI companies that often chose to develop their tools to go deep in a vertical like finance, health care, or legal—the team made an intentional choice to not assume any specific use case. Instead, as Peak said, "Let the user's needs and requests make it happen." This offered flexibility and freedom, and, as Tao noted later, "It meant we—and Manus—learned from each user's interaction and could improve our product." As they saw new use cases emerge, they handed them to the product team to fine-tune things.

As Manus took shape as an application company, relying on all different models, the team maintained close connections to the frontier labs. They guided their next iteration(s), as Peak said, "So we can ensure that the external labs are helping us."

By 2025, the team had grown to about 100 people, mostly engineering and product talent with deep consumer experience, almost all of whom were in Singapore. There were minimal sales or marketing personnel, and no research group. The founders had focused almost exclusively on the commercial user, with little interest in business-to-business markets. Aside from Tao, no one on the team had significant enterprise experience.

Internally, everyone used Manus; product managers prototyped features by asking Manus to build working mock-ups. CZ noted, "When someone complains about a tool, we ask Manus to build it." They remained very near-term focused. She added, "We choose to build for the next three months, not the next 12 months."

[h] Hallucinations referred to the mistakes or fabricated content AI could sometimes return for requests. Long-tail failures were infrequent but potentially high-impact events that could occur in the "tail" of a statistical distribution, where rare occurrences resided. Jittery sequences referred to data, video, or animation that exhibited small, rapid, random variations, often seen as unstable movement, flickering, or inconsistent timing, caused by packet delays in networks, synchronization errors in video, or intentional effects in design. Replanning instability referred to instances where minor unexpected changes in an environment or system could lead to disproportionately large and frequent revisions of an existing plan or schedule, occurring typically in automated planning, scheduling, and manufacturing systems.

The "Taste" of the Product: Evaluating and Optimizing Agent Performance

While the multi-agent architecture provided a structural foundation for quality control, Peak and the team recognized that verification was only the first step in a much broader challenge of measuring and optimizing agentic performance. As Manus transitioned from an experimental prototype to a commercial product, the team faced a critical technical challenge: how to evaluate a system that did not just talk but acted. "Evaluations are the most important thing in the field of AI," Peak noted. "Nobody has a technical moat. For an AI company, the only moat is your evaluation set, because it determines the taste of your product and guides how you iterate." To maintain this "taste," the team developed a multi-layered framework designed to balance controlled lab testing with real-world performance.

The foundation of the Manus lab-based evaluation strategy relied on automated, verifiable datasets that allowed for rapid, repeatable measurement in a controlled environment. The team prioritized three core benchmarks to serve as "regression tests," ensuring that architectural updates or new model integrations did not degrade the agent's fundamental capabilities:

- GAIA (General AI Assistants): Covered the agent's ability to handle multi-step tasks that required reasoning, tool use, and the integration of information across different modalities.
- SWE-bench**:** Evaluated software engineering capabilities by requiring the agent to resolve real-world GitHub issues, testing its capacity for complex coding, debugging, and codebase navigation.
- BrowseComp**:** Measured web navigation proficiency, specifically how effectively the agent could interact with diverse website interfaces to extract data or complete forms.

Despite the utility of these metrics, Peak remained wary of over-optimizing for them, noting that academic benchmarks often focused on complicated math or competition programming rather than the nuanced preferences of actual users.

To bridge the gap between the lab and the real world, the team turned to "field" feedback—data collected *in situ* from active users. While traditional chatbots often relied on Direct Preference Optimization (DPO)—a technique in which users chose between two static answers—the long-horizon, divergent nature of agent trajectories made this approach computationally impractical.

Instead, the team focused on real-time human feedback as a primary measurement of success. Peak discovered that users were naturally "teaching" the agent; if Manus stalled, users would intervene with specific instructions, such as suggesting a more reliable website or a specific font for a visualization. This "collective feedback" was then clustered to identify common failure patterns—such as being blocked by CAPTCHAs[i]—which guided both manual prompt "tweaks" and the development of new, atomic tools.

Beyond functional measurement, the team addressed the optimization of visual and ergonomic qualities through human subjective evaluation. Recognizing that tasks like website creation or slide generation required a human eye to judge whether the results were "visually appealing," the team employed groups of interns to manually annotate output. Peak admitted this manual process was less efficient than automated scoring but deemed it indispensable because automated reward models could

[i] CAPTCHAs, or Completely Automated Public Turing Tests to Tell Computers and Humans Apart, were security measures that relied on challenges like distorted text or image selection to verify users were human and not bots.

not yet capture the subtle aesthetic preferences of pro-sumer users. Ultimately, this framework extended to ethical responsibility: the team maintained that while AI could optimize task execution, humans remained the final arbiters of actions requiring trust or legal accountability because an agent could not take responsibility for its decisions.

Despite these advancements, the transition to a truly agentic operating core raised fundamental questions about how to bridge the gap between technical metrics and business value. How could the team make evaluation more business-relevant, moving beyond "compilation" to measure the actual return on investment (ROI) for a corporate user? Furthermore, as the agent began taking on repetitive tasks, the team debated the best approach to running A/B tests in environments that were constantly changing—such as the live web. Perhaps most importantly, the team considered how to empower users to conduct their own evaluations, ensuring the agent was not just completing tasks, but actually solved the unique, high-value problems that varied from one professional to the next. As Manus continued to scale, Peak had to decide whether to further refine these centralized lab tests or build the tools that would enable users to become the ultimate evaluators of their own AI assistants.

A Viral Launch, and Relocation

On March 5, 2025, the team posted an innocuous message on X announcing Manus's official launch. The next day, the official launch video dropped. It was not an elaborate production. Directed by Tao, it featured Peak in a corner of Butterfly Effect's Beijing office, speaking in fluent English, calmly demonstrating how Manus could take a high-level goal, break it down into steps, and execute it autonomously using a browser and other tools in a cloud-based sandbox. One observer described, "Manus was designed as a true AI agent. Give it a goal like 'analyze competitors and create a business report,' and it would deliver—browsing the web, analyzing data, generating code, planning travel, building prototypes, writing reports, all with minimal human direction."[44]

Within a week, the Manus waitlist swelled to 2 million people, on its way to 3.5 million. Discord membership rocketed to 138,000, crashing their servers and prompting the team to implement an invite-only beta system. Tao recalled, "We'd already been working around the clock, now it was for real 24/7." A month later, in April, they closed a $75 million Series B round led by Benchmark, valuing the company at approximately $500 million.[45] While the technology appeared promising, the team was a strong draw too. A Benchmark general partner said, "[The team is] one of the greatest technical teams in the world. [They have] brilliance, resilience, grit, vision, execution, and kindness."[46]

In the early summer, the team relocated its headquarters to Singapore and laid off 80 mainland employees.[47] The core engineering team all co-located in Singapore and worked closely together—often in the same room—avoiding fragmentation.

Great Product; How to Price it?

Early Core Use Cases

Initially, Manus excelled at research (i.e., deep online information gathering and summarization), slide creation (full decks), web development (full sites), data analysis, and drafting and/or sequencing tasks (i.e., write→design→structure). The team targeted consultants, solo-preneurs, analysts, and small teams. Tao said, "Mostly they were non-engineers, seeking leverage." A growing pro-sumer base began to take shape. These users brought rich behavioral signals about long-horizon tasks (slides, research, webpages). The team steered away from developing capabilities that allowed Manus to autonomously deal with health/finance-sensitive data or actions. Likewise, they steered Manus clear of payment data interactions. (See **Exhibit 7** for Manus features.)

Pricing

In December 2025, Manus was priced using a subscription-plus-credits model, in which users paid a monthly fee in exchange for a set allotment of credits that were consumed as the agent performed tasks. The free tier offered basic access at no cost, with a small daily or monthly credit allowance and occasional sign-up bonus credits, making it suitable for testing simple agent workflows. The Starter (or Basic) plan, priced at roughly $39 per month, provided 3,900 credits and supported limited concurrency, which fit solo users or light usage. The Pro plan, at around $199 per month, increased the allowance to roughly 19,900 credits, enabled more concurrent tasks, and unlocked advanced or beta features, targeting power users and small teams. For organizations, team and enterprise plans were typically priced per seat (often around $39 to $40 per user per month with a minimum team size) and included shared credit pools, collaboration and admin controls, and enterprise features such as single sign-on (SSO) and enhanced support.

The Economic Frontier: Pricing Agentic Value

As the team contemplated the future of Butterfly Effect Technology, the question of how to price Manus became as existential as its technical architecture. By late 2025, AI coding and productivity platforms had converged on several usage-based models, but these structures faced a fundamental challenge: they were often economically unsustainable. While products like Cursor and Windsurf achieved explosive growth, they frequently operated at low or even negative gross margins due to the high variable costs passed through to foundation model APIs like Claude and Gemini.

Traditionally, Manus and its competitors had relied on two primary usage-based structures to abstract these computational costs for the user:

- Direct Usage-Based Pricing**:** Users paid for exactly what they consumed on a pay-as-you-go basis, typically tied to token counts processed or specific agent runs.

- Credit Pools with Hybrid Overages**:** This model provided a fixed allocation of credits for a monthly subscription, with each agent action consuming a predetermined number of credits. Once users exhausted their allocation, they could purchase additional credit packs, a structure designed to reduce the "pain" of incremental usage.

However, these models priced "work"—the computational effort performed—rather than the actual "outcomes" achieved. This created a strategic misalignment: to reach sustainable unit economics, Manus needed to justify higher prices by increasing the value customers received. Yet, usage-based pricing disincentivized the very behavior—extensive exploration and "learning by doing"—that allowed users to climb the product learning curve and realize that value.

The leadership team wondered if outcome-based pricing might be a more sustainable alternative, particularly for autonomous background agents. Under this framework, a customer would pay the agent provider only upon successful completion of a task, such as a code refactor that passed unit tests or a web application that met a predefined specification. This framework offered strategic advantages by shifting the focus to net financial impact rather than developer-hours, while simultaneously creating internal pressure on Manus to maintain cost discipline by using lower-cost models for simple sub-tasks. Much like service-level agreements in cloud computing that shared risk through uptime guarantees, outcome-based pricing aimed to build the deep trust required for enterprise adoption. However, the team remained wary of the challenges involved in defining a "successful resolution" without creating friction with customers. For Manus, the challenge remained in choosing measurable, shared outcomes that were granular enough to be transparent but significant enough to justify the premium of an agentic solution.

Next Horizons

By December 2025, the team had opened offices in San Mateo, Tokyo, and soon Paris. Analysts estimated ZhenFund's stake in Butterfly Effect at about 10%, with Benchmark, HSG, and Tencent purportedly holding single-digit percentage stakes.[48] They projected Manus's value at more than $2 billion.[49]

Competition had intensified, with Anthropic (October 2024, December 2025), Google (November 2025), and OpenAI (March, July, and December 2025) all releasing agent frameworks. Frameworks provided the tools, structure, and environment to build, train, manage, and deploy the models, essentially acting as the assembly line for the application. Red noted, "The models contribute only about 30% to 40% of an agent's value; the rest is the engineering that is done on top of it." Indeed, agents could work 24/7, handle multi-step tasks without a human "steering" or prompting them, and work for long periods—up to seven hours in some cases—on a single task. Unlike static chatbots, these systems functioned as a model placed within a reactive environment, such as a virtual machine sandbox, allowing them to make a real-world impact by using tools, installing software, and managing complex workflows synchronously and asynchronously.

From the outset, the team had resisted pursuing the enterprise market. CZ acknowledged, "We do get a lot of requests from big enterprise companies, but in the near term, the potential of the consumer market is too huge for us to explore." PanPan added, "Many of our existing customers—pro-sumers—who are already using Manus are working in enterprise environments. Having individual employees already using Manus will help us if we go to the enterprise market." He acknowledged that Manus's existing team makeup and focus also inhibited a robust enterprise strategy. He elaborated, "We're all product guys, and we're still a very small team. Putting resources toward an enterprise product means checking compliance boxes and similar things. There's no ROI for us in that." Nonetheless, these customers—who prized reliability, auditability, and integrations—remained the most durable path to scale and margin.

Shifting toward an enterprise market, however, would require heavy investments: dedicated product and sales teams, compliance, SSO capabilities and identity integrations, audited data pipelines, sales cycles, and service teams—areas in which Butterfly Effect had few resources or capabilities, and where the frontier labs and deep-pocketed incumbents had existing positions and strong propositions (see **Exhibit 8**). In late October, Red and the cofounders had begun to prepare a team to explore the enterprise opportunity in more depth. Recent hires and the new office in San Mateo aimed to bring onboard people with enterprise-relevant experience. Red said, "I think next year the enterprise will be a very important part of the business." However, the new effort would also shift who or what product might manage Manus. Red added, "Today, the user controls Manus."

Tao looked out over the Singapore high-rises as the morning brightened. Strangely enough, his WhatsApp was lighting up with a call from a Bay Area telephone number.

Exhibit 1 The Basics of Building an AI Agent

Building an AI agent relied on using programmed rules, reinforcement learning architectures, LLM frameworks, or hybrids of several of these techniques. An agent comprised several components; a perception module that handled inputs like text, images, sensor data, or API responses; a reasoning or decision-making module that relied on rules, machine learning (ML) models, or LLMs that informed its decisions; an action module that executed actions (i.e., calling tools, updating databases, controlling hardware, or sending messages); and an environment interface that provided a way to interact with external systems (APIs, devices, other agents). It might also have a memory or state to store knowledge, past interactions, or representations of the environment.

Training an agent involved supervised learning (learning from labeled examples, i.e., predicting the next word or classifying images), reinforcement learning (RL) (learning via rewards or penalties in interactions with an environment), imitation learning (through imitations of human or other agent demonstrations); fine-tuning or instruction tuning (for LLM agents, where models were refined using human-written instructions, conversations, or preference data), or self-supervised pre-training (used for LLMs and vision models, where the model learned patterns from large sets of unlabeled data).

Source: Casewriters.

Exhibit 2 Select Agent Archetypes

Type of Agent	Description	Illustrative Products	Sample Use Cases
Analytics Agents	Agents that autonomously analyze or interpret structured datasets, generate insights, build visualizations, and assist with forecasting and modeling	Microsoft Power BI Copilot, Tableau GPT, ThoughtSpot	Monitoring key performance indicators, automated reporting, forecasting
Business Agents	Agents that autonomously execute operational workflows across enterprise systems (e.g., customer relationship management and enterprise resource planning) to automate business processes	Salesforce Agentforce, HubSpot Breeze AI Agents, Microsoft Copilot for Dynamics 365, SAP Joule	Lead qualification, customer relationship management updates, customer onboarding, marketing content generation
Conversational Agents	Dialogue-driven, rule-based AI systems focused on natural language interaction for support, assistance, and engagement	ChatGPT, Claude, Intercom Fin, Zendesk AI, Drift AI	Customer support, virtual assistants, FAQ automation, appointment scheduling, employee helpdesks
Research Agents	Agents that autonomously gather, synthesize, and analyze information from multiple sources for research and knowledge work	Perplexity AI, Elicit, ChatGPT Deep Research, Google Gemini Advanced (Research mode)	Literature reviews, market research, competitive intelligence, due diligence, source summarization
Automation/Browser Agents	Agents capable of interacting directly with web browsers or software interfaces to complete multi-step digital tasks autonomously	OpenAI Operator, Microsoft Copilot Studio, Adept ACT-1, Rabbit r1, AutoGPT	Web form submission, booking travel, user interface automation, cross-app workflows, data extraction from websites

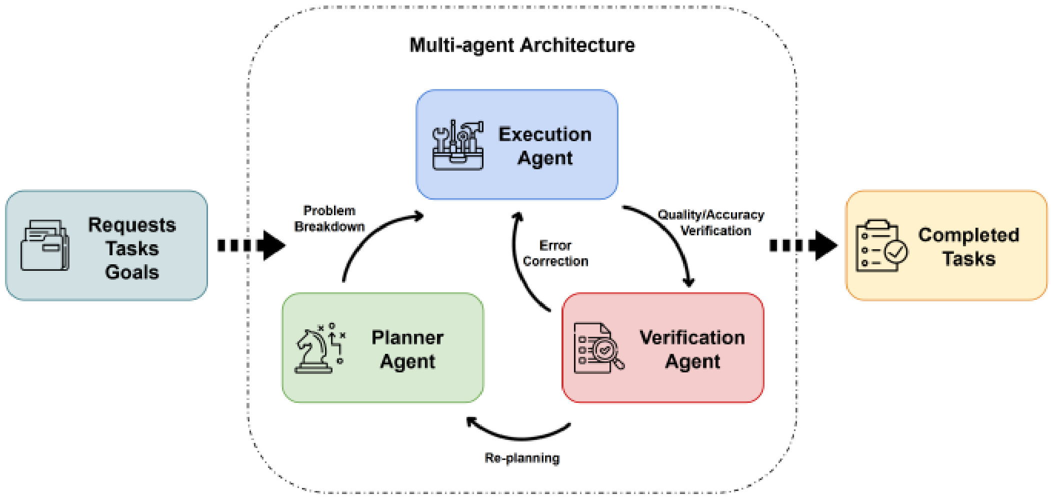
Source: Casewriters, adapted from: Aaron Holmes, "The Seven Kinds of AI Agents," The Information, July 7, 2025, https://tinyurl.com/jhuxt2ad; "What Is an AI Agent?" Google Cloud, December 4, 2025, https://cloud.google.com/discover/what-are-ai-agents; IBM, "Enterprise AI Agents: Beyond Productivity," November 21, 2025, https://www.ibm.com/think/insights/enterprise-ai-agents; DeepAI, "Conversational Agent," https://tinyurl.com/5dk5nvfp; James Watson, "AI Platform Comparison: HubSpot Breeze, Salesforce Agentforce, AWS, Google Vertex AI, Microsoft AI," Xorrox, July 3, 2025, https://tinyurl.com/24zs55vt; Perplexity AI, "What Is Perplexity?" https://tinyurl.com/n2uwsdj4; Ought, "Elicit," https://ought.org/elicit; Pendo, "AI Agent Analytics," https://www.pendo.io/glossary/ai-agent-analytics/; MindStudio, "9 AI Agents for Research and Analysis," February 6, 2026, https://www.mindstudio.ai/blog/ai-agents-research-analysis; Peakflo, "What AI Browser Agents Actually Do: Real Examples & Uses," https://blog.peakflo.co/en/agentic-workflow/ai-browser-agent; Salesforce, "Agentforce," https://www.salesforce.com/agentforce/; Tableau, "Artificial Intelligence (AI)," https://www.tableau.com/data-insights/ai; AutoGPT, "Empower Your Digital Tasks with AutoGPT," https://agpt.co/; SAP, "Joule," https://tinyurl.com/2zj4cykr; Fin, "The #1 AI Agent for All Your Customer Service," https://fin.ai/; Zendesk, "AI Built to Drive Service Resolutions," https://www.zendesk.com/service/ai/; Rabbit, https://tinyurl.com/4h43caw4; Drift, https://www.drift.ai/; all accessed March 2026.

Exhibit 3 The AI Stack: Comparisons Across Major Players, 2025

- Physical AI:
  - Wearable devices: Amazon, Apple, Google, Meta, OpenAI
  - Humanoid Robots: Amazon, Apple, Google, Meta, Nvidia, OpenAI, xAI
- Applications:
  - Enterprise AI Apps: Amazon, Anthropic, Google, Microsoft, OpenAI, xAI
  - Consumer AI Apps: Amazon, Anthropic, Apple, Google, Meta, Microsoft, OpenAI, xAI
  - Foundation large language models: Amazon, Anthropic, Apple, Google, Meta, Microsoft, Nvidia, OpenAI, xAI
- Cloud AI distribution:
  - Training clusters: Amazon, Anthropic, Apple, Google, Meta, Microsoft, Nvidia, OpenAI, xAI
  - Server rentals: Amazon, Google, Microsoft, Nvidia
  - APIs: Amazon, Anthropic, Google, Meta, Microsoft, Nvidia, OpenAI, xAI
- AI compute / server chips: Amazon, Apple, Google, Meta, Microsoft, Nvidia, OpenAI, xAI

Source: Casewriter research, adapted from Amir Efrati, "OpenAI, Meta and Their AI Rivals Ramp Up Turf Wars and Partnerships . . .," The Information, December 28, 2025; tinyurl.com/9ce4p697; "Amazon EC2 P5 instance," Amazon, tinyurl.com/y79zhbmh; "AWS Trianium," Amazon, tinyurl.com/3vxha4a6; "Amazon EC2 UltarClusters," tinyurl.com/3wxrcxap; "Amazon Bedrock," Amazon, tinyurl.com/nhc5smef; "Amazon Nova," Amazon, tinyurl.com/nhhpd464; " Amazon Q—Generative AI Assistant," Amazon, tinyurl.com/ydb8rxxa; Maria de Lourdes Zollo, "Building Bee at Amazon," Amazon News, January 5, 2026, tinyurl.com/64vj9hz3; "Amazon's next-gen AI assistant for shopping . . .," Amazon News, November 18, 2025, tinyurl.com/eejdccjn; "Think Fast, Build Faster," Claude, claude.ai/login; "Claude Enterprise," claude.com/pricing/enterprise; "Introducing the next generation of Claude," Anthropic, March 4, 2024, tinyurl.com/4trxa37v; "Start Building With Claude," Claude API Docs, tinyurl.com/mryah6xn; "AWS activates Project Rainier," Amazon News, October 29, 2025, tinyurl.com/44xrb6y2; Mark Gurman, "Apple Plots Expansion Into AI Robots, Home Security and Smart Displays," *Bloomberg*, August 13, 2025, tinyurl.com/j27xaur3; "Apple Watch," Apple, www.apple.com/watch/; "Apple Intelligence," Apple, www.apple.com/apple-intelligence/; "Apple unveils M3, M3 Pro, and M3 Max. . .," Apple, October 30, 2023, tinyurl.com/tdnk2fvt; Kif Leswing, "Apple says its AI models were trained on Google's custom chips," CNBC, July 29, 2024, tinyurl.com/2fnapta5; "Apple Intelligence Foundation Language Models," Machine Learning Research, paper, July 2024, tinyurl.com/yknup442; MacKenzie Sigalos, "Tech Google and Anthropic announce cloud deal worth tens of billions of dollars," CNBC, October 24, 2025, tinyurl.com/62ewfe8r; "Cloud Tensor Processing Units (TPUs)," Google, cloud.google.com/tpu; "AI for Every Developer," Google, ai.google.dev/; "Bring the best of Google AI to every employee, for every workflow," Google Cloud, tinyurl.com/ue2ey8p5; "intrinsic," Intrinsic, www.intrinsic.ai/; Max A. Cherney and Katie Paul, "Meta unveils plans for batch of in-house AI chips," Reuters, March 11, 2026, tinyurl.com/abnc4jsa; "Introducing the AI Research SuperCluster . . .," Meta, January 24, 2022, ai.meta.com/blog/ai-rsc/; "Bring your ideas to life with Vibes," Meta AI, ai.meta.com/meta-ai/; "Introducing the New Ray-Ban | Meta Smart Glasses," Meta, tinyurl.com/2mhmxrza; Scott Guthrie, "Maia 200. . .," Microsoft Blog, January 26, 2026, tinyurl.com/6wuhyn6b; Rani Borkar, "Microsoft Azure delivers the first large scale cluster . . .," Microsoft, October 9, 2025, tinyurl.com/4mbn83ft; "AI-power features in Windows 11," tinyurl.com/3npaw253; "Nvidia Isaac," Nvidia, developer.nvidia.com/isaac; "Nvidia Nemotron," Nvidia, tinyurl.com/4tpz3pff; "NVIDIA NIM for Developers," Nvidia, developer.nvidia.com/nim; "Nvidia DGX Cloud," Nvidia, tinyurl.com/mtbunnj7; "Nvidia DGX SuperPOD," Nvidia, tinyurl.com/6dfseftu; "H100 GPU," Nvidia, tinyurl.com/4uch9kwu; "Full Self-Driving (Supervised)," Tesla, www.tesla.com/fsd; "Welcome to xAI Documentation," xAI, docs.x.ai/overview; "Colossus," xAI, x.ai/colossus; "AI & Robotics," Tesla, hwww.tesla.com/AI; Stephen Nellis, "Startup from ex-Apple team raises $100 million, works with OpenAI," Reuters, March 8, 2023, tinyurl.com/bdfmtv2u; "ChatGPT," ChatGPT, chatgpt.com; "OpenAI Developers," OpenAI, tinyurl.com/mr4b4c7b; and Jakob Steinschaden, "Figure AI: Robot startup terminates partnership with OpenAI, relies on its own LLMs," Trending Topics, February 5, 2025, tinyurl.com/umrefffj; all accessed March 2026.

Exhibit 4 How Manus AI Works



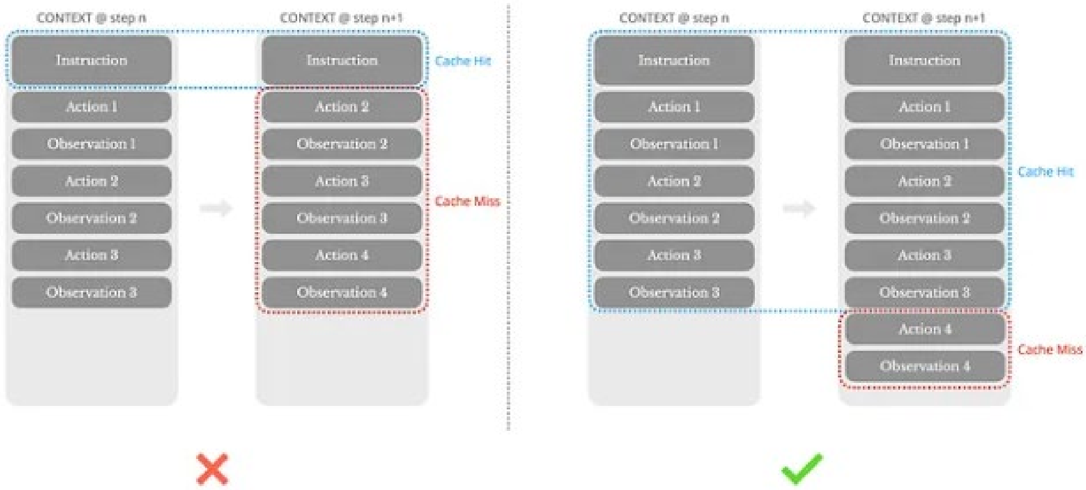
Planner Agent: This module functions as the strategist. When a user gives a request or goal, the Planner breaks the problem down into manageable sub-tasks and formulates a step-by-step plan or strategy to achieve the desired outcome.

Execution Agent: This is the action module. The Execution agent takes the Planner's plan and carries it out by invoking the necessary operations or tools. It interacts with external systems (for example, web browsers, databases, code execution environment) to gather information, perform calculations, or execute commands needed for each sub-task.

Verification Agent: Acting as quality control, this module reviews and verifies the outcomes of the Execution agent's actions. It checks results for accuracy and completeness, ensuring that each step meets the requirements before finalizing the output or moving on. The Verification agent can correct errors and trigger re-planning if needed.

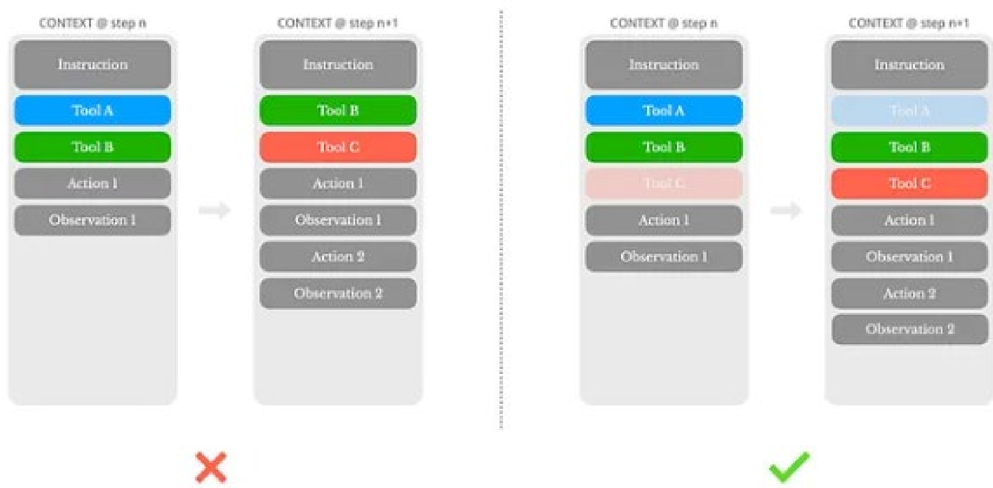
Source: Excerpted from Minjie Shen, Yanshu Li, Lulu Chen, and Qikai Yang, "From Mind to Machine: The Rise of Manus AI as a Fully Autonomous Digital Agent," arXiv:2505.02024 (Cornell University), July 20, 2025, https://arxiv.org/abs/2505.02024, accessed September 2025. Licensed under CC BY 4.0.

Exhibit 5 Design Around KV-Cache



Source: Yichao "Peak" Ji, "Context Engineering for AI Agents: Lessons from Building Manus," Manus, https://manus.im/blog/Context-Engineering-for-AI-Agents-Lessons-from-Building-Manus, accessed January 2026.

Exhibit 6 "Mask, Don't Remove"



Source: Yichao "Peak" Ji, "Context Engineering for AI Agents: Lessons from Building Manus," Manus, https://manus.im/blog/Context-Engineering-for-AI-Agents-Lessons-from-Building-Manus, accessed January 2026.

Exhibit 7 Manus Features

Autonomous Task Execution: Manus AI could carry out complex sequences of actions with minimal user intervention. Once given a high-level task, it would plan, execute, and finalize the task largely on its own. In 2025, this went far beyond the typical AI, which required the user to break down the problem or confirm each step. As the Manus team noted, “It excels at various tasks in work and life, getting everything done while you rest.” It could generate a detailed report with visuals and texts from raw entirely autonomously, for example, or perform all the steps of booking a trip after a user requested a vacation plan.

Multi-Modal Understanding: Manus AI was designed to process and generate multiple types of data, including text (e.g., generating reports, answering queries), images (e.g., analyzing visual content), and code (e.g., automating programming tasks). Its versatility meant Manus could tackle tasks like reading a diagram or X-ray and writing an explanation of it or debugging a piece of software based on both the code and error screenshots.

Advanced Tool Use: Manus AI was adept at integrating with external tools and software applications to augment its abilities. It has built-in support for web browsing, so it can fetch up-to-the-minute information from the Internet. It could interface with productivity software (i.e., creating or editing spreadsheets and documents) and query databases. Its ability to interact with external applications made Manus AI an ideal tool for businesses looking to automate workflows. Integrating tool use into an AI agent was challenging, and Manus’s effective tool usage was a major innovation in bridging AI with practical automation tasks.

Continuous Learning and Adaptation: Manus AI learned continuously from user interactions and optimized its processes to provide personalized and efficient responses. This ensured that over time, the AI became more tailored to the specific needs of the user. If a user consistently preferred data presented in a certain format or tone, Manus adapted to those preferences in future outputs. This adaptive learning happened during use, complementing its initial offline training. Additionally, the Manus team emphasized ethical safeguards and transparency, meaning the system was designed to adjust its actions to avoid unsafe outcomes and to align with human intentions as it gained experience.

Source: Excerpted and adapted from Minjie Shen, Yanshu Li, Lulu Chen, and Qikai Yang, “From Mind to Machine: The Rise of Manus AI as a Fully Autonomous Digital Agent,” arXiv:2505.02024 (Cornell University), July 20, 2025, https://arxiv.org/abs/2505.02024, accessed September 2025. Licensed under CC BY 4.0.

Exhibit 8 AI Feature Comparisons: Anthropic, Google, Monica (Manus), OpenAI

Feature	Computer Use (Anthropic)	Mariner (Google)	Manus AI (Monica)	Operator (OpenAI)
Agent Type	Browser-based	Browser-based (Chrome extension)	Browser-based (operates in Linux sandboxes)	Browser-based
Autonomous web browsing	Yes*	Yes	Yes	Yes
Form filling and data entry	Yes*	Yes	Yes	Yes
Online shopping and reservations	Yes*	Yes	Yes	Yes
Multi-modal input/output (text, images)	Limited*	Yes	Yes	Limited
Integration with external APIs	Yes	N/A	No	No
Availability	Beta (API access)	Research Phases	Beta (invite-only)	Subscribers

Source: Adapted from Minjie Shen, Yanshu Li, Lulu Chen, and Qikai Yang, "From Mind to Machine: The Rise of Manus AI as a Fully Autonomous Digital Agent," arXiv:2505.02024 (Cornell University), July 20, 2025, https://arxiv.org/abs/2505.02024, accessed September 2025. Licensed under CC BY 4.0.

Note: * = Requires integration through the API.

Endnotes

[1] Juro Osawa, "Meta's Acquisition Values Manus at More Than $2 Billion," The Information, December 30, 2025, https://www.theinformation.com/briefings/metas-acquisition-values-manus-2-billion, accessed December 2025.

[2] Eric Schmidt, cited in Tony Peng, "Meta's Manus Acquisition: A New Playbook for Chinese-Founded AI Startups," Recode China AI, January 1, 2026, https://recodechinaai.substack.com/p/metas-manus-acquisition-a-new-playbook, accessed January 2026.

[3] Aaron Holmes, "The Seven Kinds of AI Agents," The Information, July 7, 2025, https://www.theinformation.com/articles/seven-kinds-ai-agents, accessed December 2025.

[4] Holmes, "The Seven Kinds of AI Agents."

[5] Holmes, "The Seven Kinds of AI Agents."

[6] Holmes, "The Seven Kinds of AI Agents."

[7] Holmes, "The Seven Kinds of AI Agents."

[8] Holmes, "The Seven Kinds of AI Agents."

[9] Holmes, "The Seven Kinds of AI Agents."

[10] Holmes, "The Seven Kinds of AI Agents."

[11] Holmes, "The Seven Kinds of AI Agents."

[12] Aaron Holmes, "A Reality Check on Agents," The Information, October 21, 2025, https://www.theinformation.com/articles/reality-check-agents, accessed December 2025.

[13] Holmes, "The Seven Kinds of AI Agents."

[14] Holmes, "The Seven Kinds of AI Agents."

[15] Holmes, "The Seven Kinds of AI Agents."

[16] Holmes, "The Seven Kinds of AI Agents."

[17] Holmes, "The Seven Kinds of AI Agents."

[18] Holmes, "The Seven Kinds of AI Agents."

[19] "Cursor Secures $2.3 Billion Series D Financing at $29.3 Billion Valuation to Redefine How Software is Written," BusinessWire, November 13, 2025, https://www.businesswire.com/news/home/20251113939996/en/Cursor-Secures-%242.3-Billion-Series-D-Financing-at-%2429.3-Billion-Valuation-to-Redefine-How-Software-is-Written, accessed December 2025.

[20] Suhasini Srinivasaragavan, "Google hires Windsurf heads in $2.4bn licensing deal," Silicon Republic, July 14, 2025, https://www.siliconrepublic.com/business/google-windsurf-license-varun-mohan-douglas-chen, accessed December 2025.

[21] Mike Isaac, "Cognition Buys Windsurf as A.I. Frenzy Escalates," *The New York Times,* July 14, 2025, https://www.nytimes.com/2025/07/14/technology/cognition-ai-windsurf.html, accessed December 2025.

[22] Isaac, "Cognition Buys Windsurf as A.I. Frenzy Escalates."

[23] Erin Griffith and Cade Metz, "The New A.I. Deal: Buy Everything but the Company," *The New York Times,* August 8, 2024, https://www.nytimes.com/2024/08/08/technology/ai-start-ups-google-microsoft-amazon.html, accessed December 2025.

[24] Griffith and Metz, "The New A.I. Deal: Buy Everything but the Company."

[25] Griffith and Metz, "The New A.I. Deal: Buy Everything but the Company."

[26] Holmes, "A Reality Check on Agents."

[27] Holmes, "A Reality Check on Agents."

28 Holmes, "A Reality Check on Agents."

29 Shikhar Gupta, "What is Manus, the Singapore-based AI startup acquired for US$2 billion by Meta?" The Business Times, December 30, 2025, https://www.businesstimes.com.sg/international/asean/what-manus-singapore-based-ai-startup-acquired-us2-billion-meta, accessed December 2025.

30 Holmes, "A Reality Check on Agents."

31 Peng, "Meta's Manus Acquisition: A New Playbook for Chinese-Founded AI Startups."

32 Gupta, "What is Manus, the Singapore-based AI startup acquired for US$2 billion by Meta?"

33 Russell Flannery, "Forbes China 30 Under 30 List: Ji Yichao's Mammoth Ambitions," *Forbes*, March 20, 2013, https://www.forbes.com/sites/russellflannery/2013/03/19/forbes-china-30-under-30-list-ji-yichaos-mammoth-ambitions/, accessed December 2025.

34 "Innovators Under 35," *MIT Technology Review*, 2025, https://www.innovatorsunder35.com/the-list/yichao-peak-ji/, accessed December 2025.

35 Yichao "Peak" Ji, "Context Engineering for AI Agents: Lessons from Building Manus," Medium, July 18, 2025, https://medium.com/@peakji/context-engineering-for-ai-agents-lessons-from-building-manus-71883f0a67f2, accessed December 2025.

36 "Innovators Under 35," MIT Technology Review, 2025.

37 "Atlassian Completes Acquisition of The Browser Company of New York," Businesswire, October 21, 2025, https://www.businesswire.com/news/home/20251021473486/en/Atlassian-Completes-Acquisition-of-The-Browser-Company-of-New-York, accessed December 2025.

38 Ji, "Context Engineering for AI Agents: Lessons from Building Manus."

39 Ji, "Context Engineering for AI Agents: Lessons from Building Manus."

40 Ji, "Context Engineering for AI Agents: Lessons from Building Manus."

41 Ji, "Context Engineering for AI Agents: Lessons from Building Manus."

42 Minjie Shen, Yanshu Li, Lulu Chen, and Qikai Yang, "From Mind to Machine: The Rise of Manus AI as a Fully Autonomous Digital Agent," arXiv:2505.02024 (Cornell University), July 20, 2025, https://arxiv.org/abs/2505.02024, accessed September 2025.

43 Shen, Li, Chen, and Yang, "From Mind to Machine: The Rise of Manus AI as a Fully Autonomous Digital Agent."

44 Peng, "Meta's Manus Acquisition: A New Playbook for Chinese-Founded AI Startups."

45 Lee Chong Ming, "All You Need to Know About Manus, the Startup Meta is Acquiring," Business Insider, December 31, 2025, https://www.businessinsider.com/what-is-manus-ai-meta-acquisition-chinese-startup-singapore-agent-2025-12, accessed December 2025.

46 Peng, "Meta's Manus Acquisition: A New Playbook for Chinese-Founded AI Startups."

47 Peng, "Meta's Manus Acquisition: A New Playbook for Chinese-Founded AI Startups."

48 Osawa, "Meta's Acquisition Values Manus at More Than $2 Billion."

49 Osawa, "Meta's Acquisition Values Manus at More Than $2 Billion."